

COSC-1436 Programming Fundamentals I

TARRANT COUNTY COLLEGE DISTRICT MASTER SYLLABUS

COURSE DESCRIPTION

This course introduces the fundamental concepts of structured programming, and provides a comprehensive introduction to programming for computer science and technology majors. Topics include software development methodology, data types, control structures, functions, arrays, and the mechanics of running, testing, and debugging. This course assumes computer literacy. (This course is included in the Field of Study Curriculum for Computer Science.)

Recommended Prerequisite: COSC-1301

COURSE TYPE

Academic Non-Core

COURSE GOALS AND LEARNING OUTCOMES

Upon successful completion of this course, students will:

1. Describe how data are represented, manipulated, and stored in a computer.
2. Categorize different programming languages and their uses.
3. Understand and use the fundamental concepts of data types, structured programming, algorithmic design, and user interface design.
4. Demonstrate a fundamental understanding of software development methodologies, including modular design, pseudo code, flowcharting, structure charts, data types, control structures, functions, and arrays.
5. Develop projects that utilize logical algorithms from specifications and requirements statements.
6. Demonstrate appropriate design, coding, testing, and documenting of computer programs that implement project specifications and requirements.
7. Apply computer programming concepts to new problems or situations.

COURSE ASSESSMENT

Student success is measured by a variety of assessment techniques aligned with course goals and learning outcomes. Individual faculty members are responsible for designing evaluation instruments to measure student mastery of course goals and

learning outcomes and indicating the nature of such evaluation instruments in the instructor's class requirements.

STUDENT ACCESSIBILITY RESOURCES

Any student with a documented disability needing academic accommodations is required to contact the Student Accessibility Resources (SAR) Office located on each campus to schedule an appointment with the Coordinator of SAR. All discussions are confidential. Because SAR accommodations may require early planning and are not provided retroactively, students are encouraged to contact SAR as early in the semester as possible. SAR is responsible for approving and coordinating all disability-related services. TCC professors will honor requests for accommodation when they are issued by SAR.

SCHOLASTIC DISHONESTY

Students are responsible for adhering to the TCCD policy on scholastic dishonesty as stated in the online student handbook at www.tccd.edu.

RESOURCES

For additional information regarding the student handbook, academic calendar, course evaluations, attendance policy, SCANS skills, Core Competencies, etc., see www.tccd.edu.